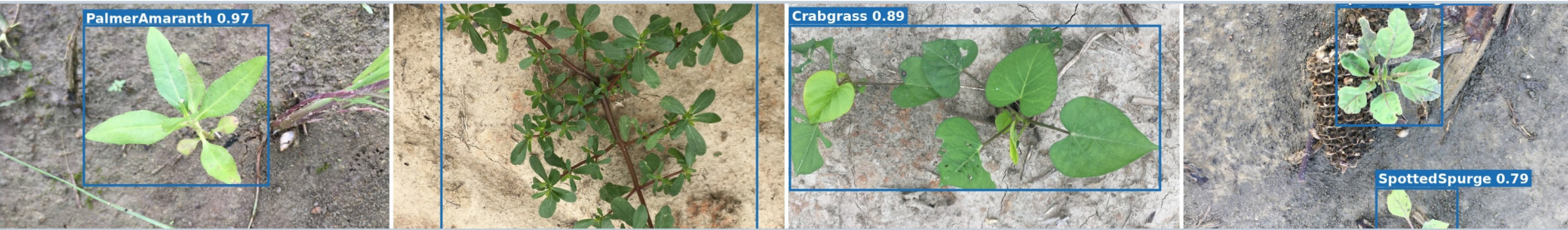


Every robot the MTSU Great Robotics Lab owns reports into one platform. Agents on it harvest, filter, label and train; a third watches those two. This is what six months of running it taught us.

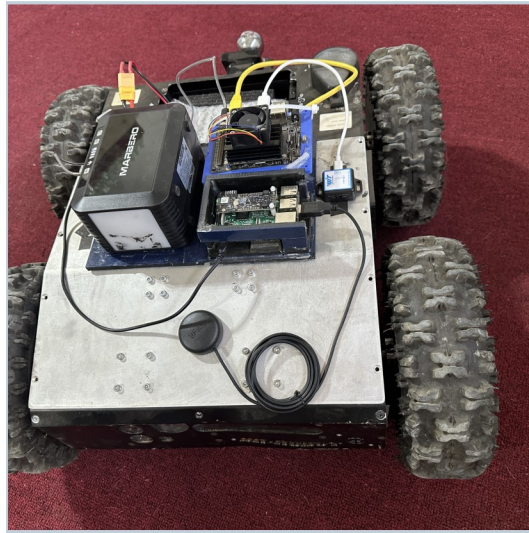
WHAT EACH PROJECT IS, AND WHAT IT IS FOR

Weed Detection



Two agents work the corpus unasked: one harvests imagery and audits it, the other trains a detector small enough to ride the robots. More than 150,000 labelled images.

Rover



A tracked rover narrow enough to turn inside a row and reach where a tractor cannot. Camera, GPS and two inertial units stream live; the stream trains a driving model.

Laser Weed Control Robot



A twin-laser weeder. A YOLO detector finds the weed; two galvanometer lasers aim and fire, and it dies where it stands. No chemical, nothing touches the crop beside it.

Humanoid and Quadruped



A humanoid and a quadruped, out on the farm to record where wheels do not go. Their project takes the ingest contract the Rover uses; any session they open is governed.

Figure 1. Every project on the platform opened the same way, same ingest contract, same agents train and evaluate it.

HOW A ROUND RUNS, AND WHAT IT DOES WHEN IT IS STUCK

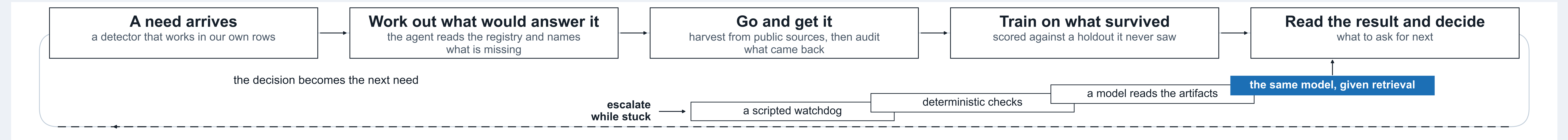


Figure 2. A need arrives, the agent works out what data would answer it, fetches it, trains on what survives the audit, scores, decides what to ask next. Stuck, it climbs the ladder. Dashed: the decision is recorded, nothing reads it back.

ONE PLATFORM TAKES ANY ROBOT, ANY DATASET

- agentAI is one platform for every lab robot
A robot, drone or sensor rig is a project with its own keys, people and locked dataset registry. Agents harvest, filter, label and train against it. Adding a robot is a configuration change.
- The Rover and the laser robot both report in
The Rover runs on tracks and streams camera, GPS and two IMUs. The Laser Weed Control Robot carries a galvanometer laser, a down camera over the work zone and a forward camera.
- A drive becomes a dataset as it opens
The platform registers a vehicle's session as a governed dataset while the drive runs. Frames and sensor batches arrive over one API key. The laser robot sends no position.
- Any data type, any field, one project each
Open a project, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the Rover and the laser robot.

THE BRAIN GETS STUCK AND ASKS UPWARD

Grounded recall separates the reviewers

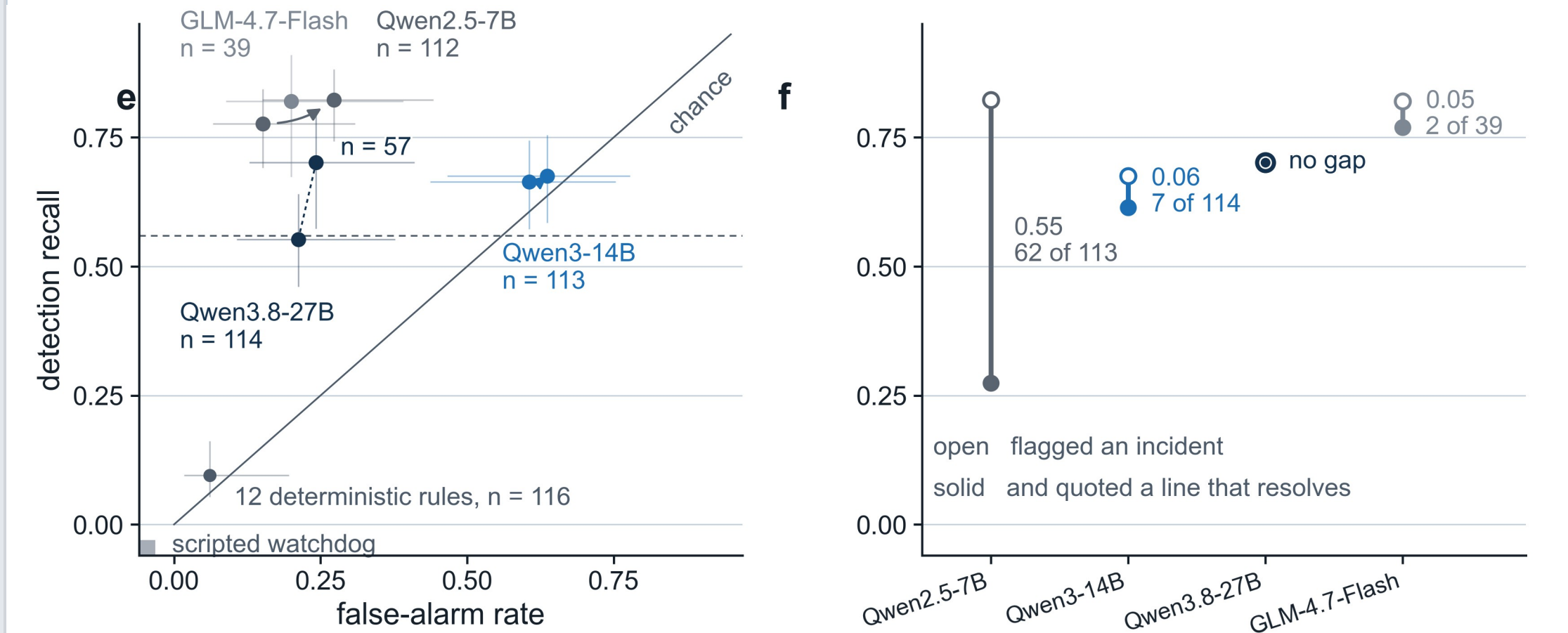


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from our own record, scored by our own scorer. Tail: model on raw artifacts; head: same model with retrieval. Dotted arrow: ends scored on different cases.

Escalating is worth it, and the reason is grounding rather than size: retrieval is worth +0.149 recall to the 27 B and +0.012 to the 14 B. This part is built and is switched off today.

Where a model runs decides whether its verdict counts
A router maps a role -- collector, judge, reviewer, planner -- onto a model and a place, and the place decides whether the verdict counts. Deciding work runs on the cluster.

AGENTS RAN THE LOOP, NOBODY IN THE ROOM

Fifteen rounds ran with nobody in the room
Two cluster jobs, one collecting and one training, ran against the locked registry. Every round reported success. This part is built and is switched off today.

MORE DATA STOPPED HELPING

More harvested data cost accuracy; we measured it

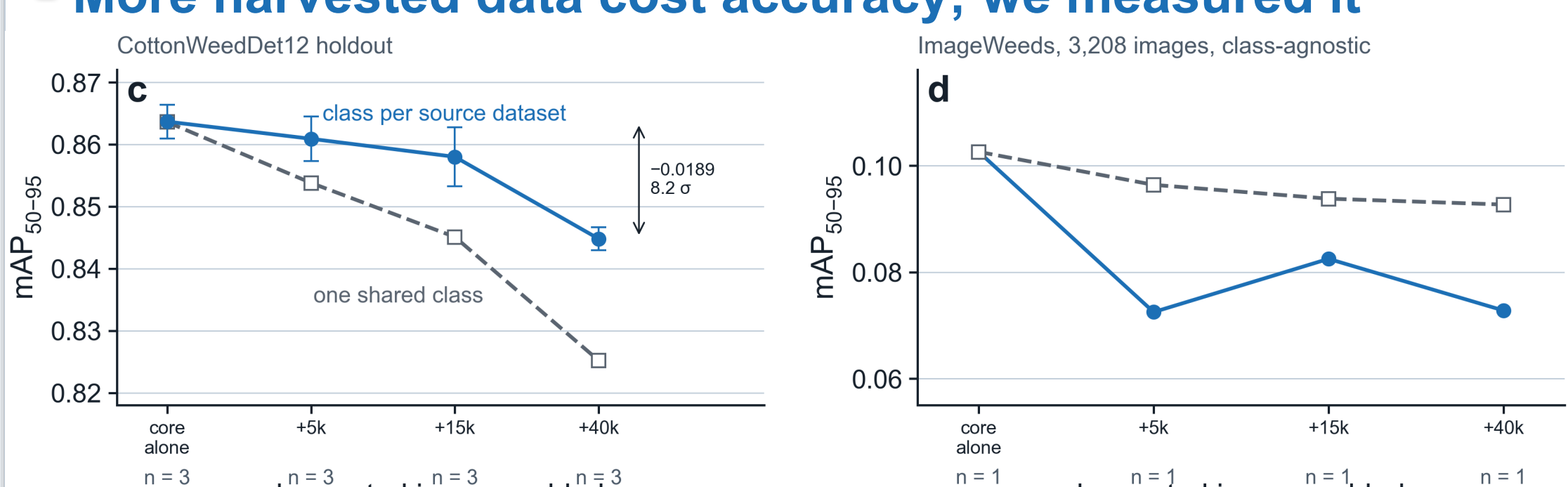


Figure 4. (c) Three seeds per rung, mean and 1 s.d., on our sealed holdout. (d) Same eight checkpoints, unseen weed dataset, one seed per rung. Both exams fall.

Starting fresh instead of from the last round's weights recovers +0.0287, 5.1 times this recipe's seed spread. That is the change the next campaign makes.

MORE DATA STOPPED HELPING (CONT.)

- The recipe was not why more data stopped helping
More data arrived; the detector did not improve. Augmentation, replaying known species, completing labels: each left the same gap. What stood was our auto-labeller's boxes.
- A vision model can judge a dataset; a chat model cannot
Professor Zhang's rule: discard what sits too far from data we trust. Shown a candidate box, Gemma 4 rejects none of the false ones, so the test is DINOv2 geometry. Fifty-one datasets out, retrained, accuracy did not come back.
- The unit of curation has to match the unit of noise
An auto-label error is one wrong box in an otherwise fine image, so a dataset-level filter cannot see it. The per-box tools are written, not re-measured. That is the open front.
- A brain's falsifiable hypothesis, and it was wrong
Harvested boxes carry a class taken from the website they came from. A brain blamed that, wrote its falsifying condition first, and a human approved. Four runs falsified it. The real cost was density: nine boxes an image against under two.

WHERE THIS GOES: ONE PLATFORM, ANY DATASET, ANY ROBOT

Agents will run unasked, iterate on their own output, escalate to a larger model when out of their depth, all inside a sandbox. Every capability stays in the platform after its experiment, so the next student inherits it working. This is the direction, not a description of what runs today.